The Parallel Postulate

If you look on our postulate sheet, you will see three different version of Postulate 5.

(Euclid) If a given line crosses two other lines such that the interior angles on the same side of the given line are less than two right angles, then the two other lines, if extended indefinitely, will meet on side of the given line where the angles are less than two right angles.
- (Playfair) For any given line and a given point that is not on the line, there exists one and only one line through the given point and parallel to the given line.
- (Wu) Given a line L and point P not on L, then through P passes at most one line that does not intersect L. (Note that this is a little weaker than the other two because it does not imply existence)

The first one is Euclid's as he stated it. It actually has two parts, the fact that the two lines will meet and the fact that they will meet on a particular side of the transversal. It's actually the fact that the lines will meet that is the most important, but we'll look at both parts.

The second one is named for John Playfair, a Scottish mathematician who published a well-known version of Euclid's Elements in 1795. It is the most common version of the postulate in English language geometry books.

The last one is a slightly weaker version that I got from a book by Hung-His Wu, a modern mathematician currently at UC Berkeley. Wu's version of the postulate leaves out the existence of a parallel line, since that is fairly easy to prove using Theorem 25.

In this assignment, we want to prove that Euclid's version and Playfair's version are equivalent. This means that if I assume either one, then I can prove the other. Hint: When you are doing this, you are probably going to want to use the fact that we can construct an angle equal to a given angle, along with Theorem 25, to produce a parallel line. Then you work from there.

1. Assume Euclid, prove Playfair.

Assume that you have added Euclid's version of the Parallel Postulate to your list of Postulates. Now we need to prove Playfair's. Setup is something like this:

Given line l and a point P not on l,

   a. Prove that there exists a line through point P parallel to line l, (This should be fairly easy)
   b. Show that any other line through P will not be parallel to l.

2. Assume Playfair, prove the first part of Euclid (the fact that the lines will meet).

Assume that you have added Playfairs's version of the Parallel Postulate to your list of Postulates. Now we need to prove Euclid's. Setup is something like this:

Given two lines *l* and *m* cut by a transversal such that the sum of interior angles on one side of the transversal is less than two right angles (i.e. less than 180 degrees).

   a. Prove that lines *l* and *m* will intersect.

(You'll want to make a drawing and name some points, so your setup will be a little clearer than mine.)

3. Prove the second part of Euclid's Postulate.

Here we want to prove that the two lines will intersect on the side of the transversal that has the two angles that add up to less than 180 degrees. You can use the setup you used to prove the first part of Euclid's postulate and you already proved that the lines intersect somewhere. To do this, you will need the Separation Postulates (look on the Postulates sheet). You will probably need a little lemma about angle measure.

Lemma: If angle ABC and angle ABD share side AB with C and D on the same side of AB and angle ABC less than angle ABD, then point C is on the interior of angle ABD. (This means, by definition, that point C is on the same side of line AB as point D, and on the same side of line BD as point A.)